$$= \sqrt{2\times3\times7\times7\times2\times3\times3\times3\times1}$$

$$= 2\times3\times7\times3$$

$$= 126 \ m^2$$

Given the cost of levelling is ₹ 20 per m^2.

The total cost of levelling the field $= 20\times126 =$ ₹ 2520.

Example 7.2

Three different triangular plots are available for sale in a locality. Each plot has a perimeter of 120 m. The side lengths are also given:

Shape of plot	Perimeter	Length of sides
Right angled triangle	120 m	30 m, 40 m, 50 m
Acute angled triangle	120 m	35 m, 40 m, 45 m
Equilateral triangle	120 m	40 m, 40 m, 40 m

Help the buyer to decide which among these will be more spacious.

Solution

For clarity, let us draw a rough figure indicating the measurements:

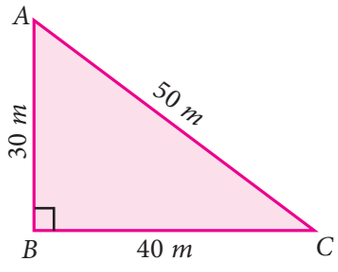


Fig. 7.4

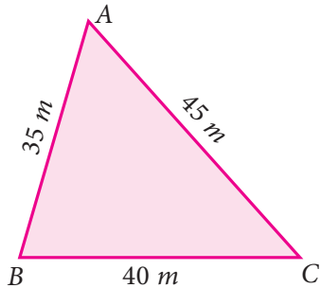


Fig. 7.5

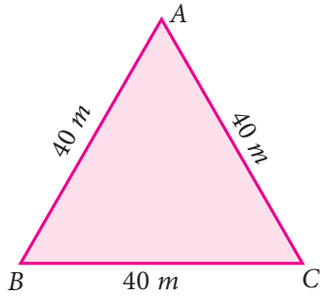


Fig. 7.6

(i) The semi-perimeter of Fig.7.4, $s = \dfrac{30+40+50}{2} = 60 \ m$

Fig.7.5, $s = \dfrac{35+40+45}{2} = 60 \ m$

Fig.7.6, $s = \dfrac{40+40+40}{2} = 60 \ m$

Note that all the semi-perimeters are equal.

(ii) Area of triangle using Heron's formula:

In Fig.7.4, Area of triangle $= \sqrt{60(60-30)(60-40)(60-50)}$

$$= \sqrt{60\times30\times20\times10}$$

7 Mensuration.indd 250 26-12-2019 14:16:21